Name: _____

TOPIC TEST 1

Chapter 2: Complex numbers and De Moivre's theorem

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks).
- Part B: 6 short-answer questions (12 marks).
- Part C: 2 analysis questions (7 marks).

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 The value of $i + i^2 + i^3 + ... + i^{78} + i^{79}$ is

A i

B 1

C -1

D $-i$

E 0

[1 mark]

2 $\frac{\sqrt{3}}{2} - \frac{i}{2}$ is equal to

A $2\left[\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right]$

B $\left[\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right]$

C $2\left[\cos\left(\frac{-\pi}{6}\right) + i\sin\left(\frac{-\pi}{6}\right)\right]$

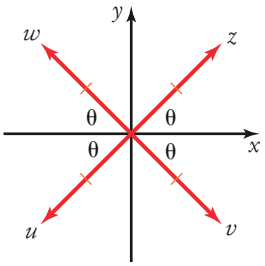
D $\left[\cos\left(\frac{-\pi}{6}\right) + i\sin\left(\frac{-\pi}{6}\right)\right]$

E $\left[\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right]$

[1 mark]

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3 In the diagram, the complex numbers z, w, u and v are shown.



Which of the following is true?

A $v = \overline{z}$

B $v = -\overline{z}$

C $w = \overline{z}$

D $u = -\overline{z}$

E $v = -z$

[1 mark]

4 The complex number $\dfrac{1+i}{1-i}$ is equivalent to

A i

B $-i$

C 1

D -1

E $\sqrt{2}$

[1 mark]

5 The complex number $\left[i\cos\left(\dfrac{\pi}{3}\right) - \sin\left(\dfrac{\pi}{3}\right)\right]$ in mod-arg form is:

A $\left[\cos\left(\dfrac{\pi}{3}\right) + i\sin\left(\dfrac{\pi}{3}\right)\right]$

B $\left[\cos\left(\dfrac{-\pi}{3}\right) + i\sin\left(\dfrac{-\pi}{3}\right)\right]$

C $\left[\cos\left(\dfrac{\pi}{3}\right) - i\sin\left(\dfrac{\pi}{3}\right)\right]$

D $\left[\cos\left(\dfrac{5\pi}{6}\right) + i\sin\left(\dfrac{5\pi}{6}\right)\right]$

E $\left[\cos\left(\dfrac{5\pi}{6}\right) - i\sin\left(\dfrac{5\pi}{6}\right)\right]$

[1 mark]

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Part B

Short-answer questions

6 short-answer questions
12 marks
Show your working where appropriate.

1 Simplify $(1 + 4i)(3 - 5i)$.

[2 marks]

2 Express $\dfrac{1}{2 - 3i}$ in the form $x + yi$.

[2 marks]

3 Calculate $|z|$ if $z = \dfrac{5 - i\sqrt{3}}{2}$.

[2 marks]

4 Use De Moivre's theorem to simplify $\left[\cos(3\theta) + i\sin(3\theta)\right]^4$.

[1 mark]

5 Simplify $\dfrac{\cos\left(\dfrac{2\pi}{3}\right) + i\sin\left(\dfrac{2\pi}{3}\right)}{\cos\left(\dfrac{-2\pi}{3}\right) + i\sin\left(\dfrac{-2\pi}{3}\right)}$.

[2 marks]

6 Express $\left[\cos\left(\dfrac{5\pi}{12}\right) + i\sin\left(\dfrac{5\pi}{12}\right)\right]\left[\cos\left(\dfrac{\pi}{3}\right) + i\sin\left(\dfrac{\pi}{3}\right)\right]$ in the form $x + yi$.

[3 marks]

Part C

Analysis questions

2 analysis questions
7 marks
Show your working where appropriate.

1 Show that $\dfrac{1}{\left(1 - i\sqrt{3}\right)^6}$ is always real.

[4 marks]

2 Prove for complex z that $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + z_1\overline{z}_2 + \overline{z}_1 z_2$.

[3 marks]

[Total marks: 24]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $i + i^2 + i^3 + \ldots + i^{78} + i^{79} = i - 1 - i + 1 + \ldots + i - 1 - i = -1$

$\therefore$ **C** **[1 mark]**

2 $\overline{z}$ is the reflection of z over the x-axis.

$\therefore$ **A** **[1 mark]**

3 $r = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1,\ \cos(\theta) = \frac{\sqrt{3}}{2},\ \sin(\theta) = \frac{-1}{2} \Rightarrow \theta = \frac{-\pi}{6}$

$\therefore$ **D** **[1 mark]**

4 $\frac{1+i}{1-i} = \frac{1+i}{1-i} \times \frac{1+i}{1+i} = \frac{1+2i-1}{1+1} = i$

$\therefore$ **A** **[1 mark]**

5 $\left[i\cos\left(\frac{\pi}{3}\right) - \sin\left(\frac{\pi}{3}\right)\right] = i\cos\left(\frac{-\pi}{3}\right) + \sin\left(\frac{-\pi}{3}\right)$

$= i\sin\left(\frac{\pi}{2} - \frac{-\pi}{3}\right) - \cos\left(\frac{\pi}{2} - \frac{-\pi}{3}\right)$

$= i\sin\left(\frac{5\pi}{6}\right) + \cos\left(\frac{5\pi}{6}\right)$

$\therefore$ **D** **[1 mark]**

Part B – Short-answer questions

1 $(1 + 4i)(3 - 5i) = 3 + 7i + 20$ **[1 mark]**

$= 23 + 7i$ **[1 mark]**

[2 marks altogether]

2 $|z| = \sqrt{\frac{25}{4} + \frac{3}{4}}$ **[1 mark]**

$= \sqrt{\frac{28}{4}} = \sqrt{7}$ **[1 mark]**

[2 marks altogether]

3 $\frac{1}{2-3i} = \frac{1}{2-3i} \times \frac{2+3i}{2+3i} = \frac{2+3i}{4+9}$ **[1 mark]**

$= \frac{2}{13} + \frac{3}{13}i$ **[1 mark]**

[2 marks altogether]

4 $\left[\cos(3\theta) + i\sin(3\theta)\right]^4 = \cos(12\theta) + i\sin(12\theta)$ **[1 mark]**

5 $\dfrac{\cos\left(\frac{2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}\right)}{\cos\left(\frac{-2\pi}{3}\right)+i\sin\left(\frac{-2\pi}{3}\right)}$

$= \cos\left(\frac{2\pi}{3}-\frac{-2\pi}{3}\right)+i\sin\left(\frac{2\pi}{3}-\frac{-2\pi}{3}\right)$ **[1 mark]**

$= \cos\left(\frac{4\pi}{3}\right)+i\sin\left(\frac{4\pi}{3}\right)$

$= \cos\left(\frac{-2\pi}{3}\right)+i\sin\left(\frac{-2\pi}{3}\right)$ **[1 mark]**

[2 marks altogether]

6 $\left[\cos\left(\frac{5\pi}{12}\right)+i\sin\left(\frac{5\pi}{12}\right)\right]\left[\cos\left(\frac{\pi}{3}\right)+i\sin\left(\frac{\pi}{3}\right)\right]$

$= \left[\cos\left(\frac{5\pi}{12}+\frac{4\pi}{12}\right)+i\sin\left(\frac{5\pi}{12}+\frac{4\pi}{12}\right)\right]$ **[1 mark]**

$= \cos\left(\frac{3\pi}{4}\right)+i\sin\left(\frac{3\pi}{4}\right)$ **[1 mark]**

$= -\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}i$ **[1 mark]**

[3 marks altogether]

Part C – Analysis questions

1 $r=\sqrt{1+3}=2,\ \theta=-\frac{\pi}{3}$ **[1 mark]**

$\dfrac{1}{\left(1-i\sqrt{3}\right)^6}=\dfrac{1}{\left[2\,\mathrm{cis}\left(\frac{-\pi}{3}\right)\right]^6}=\dfrac{1}{64\,\mathrm{cis}(-2\pi)}$ **[1 mark]**

$= \frac{1}{64}\mathrm{cis}(2\pi)$ **[1 mark]**

$= \frac{1}{64}\left[\cos(2\pi)+i\sin(2\pi)\right]=\frac{1}{64}$, which is real. **[1 mark]**

[4 marks altogether]

2 Let z be a complex number.

$\text{LHS} = |z_1 + z_2|^2$

$= (z_1 + z_2)(\overline{z_1 + z_2})$ using the identity $|w|^2 = w\overline{w}$ **[1 mark]**

$= (z_1 + z_2)(\overline{z}_1 + \overline{z}_2)$

$= z_1\overline{z}_1 + z_2\overline{z}_2 + z_1\overline{z}_2 + \overline{z}_1 z_2$ **[1 mark]**

$= |z_1|^2 + |z_2|^2 + z_1\overline{z}_2 + \overline{z}_1 z_2$

$=$ RHS using the identity $|w|^2 = w\overline{w}$ QED **[1 mark]**

[3 marks altogether]

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